Appendix A. Study area details and environmental data sources

In this document we provide details on the study area and sources of environmental data used to analyze the cumulative impacts of a mine and wind power plants on reindeer in the Malå Sámi reindeer herding community (RHC) in Northern Sweden. More specifically, we: (i) detail the definition of the seasons in each year for which we used GPS data; (ii) describe all the environmental data used in the study, pointing to sources, geographical information, thematic resolution, and details on the processing and preparation of the data layers; (iii) present a summary statistics of the values for the environmental variables for the study area.

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1 Seasons

Table A1 shows the dates of start and end of each season for each year, defined based on information from the reindeer herders in the Malå RHC.

Year	Phase	Calving	Summer	Autumn
2008	Before construction	2008-05-12 to 2008-06-18	2008-07-15 to 2008-09-15	2008-09-16 to 2008-11-18
2009	Before construction	2009-05-02 to 2009-06-19	2009-07-14 to 2009-09-15	2009-09-16 to 2009-11-07

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Year	Phase	Calving	Summer	Autumn
2010	Construction	2010-05-10 to 2010-06-24	2010-07-14 to 2010-09-15	2010-09-16 to 2010-11-05
2011	Construction	2011-05-10 to 2011-06-22	2011-07-09 to 2011-09-15	2011-09-16 to 2011-11-30
2015	Operation	2015-05-01 to 2015-06-25	2015-07-15 to 2015-09-15	2015-09-16 to 2015-11-13
2016	Operation	2016-05-05 to 2016-06-23	2016-07-15 to 2016-09-15	2016-09-16 to 2016-11-30
2017	Operation	2017-05-03 to 2017-06-28	2017-07-08 to 2017-09-15	2017-09-16 to 2017-11-15
2018	Operation	2018-05-05 to 2018-06-26	2018-07-12 to 2018-09-15	2018-09-16 to 2018-11-30

Table A1. Dates of the calving, summer, and autumn seasons in each of the years of reindeer monitoring.

2 Environmental data

All environmental data was processed in GRASS GIS environment, version 7.8 (GRASS Development Team, 2017), in SWEREF99 (EPSG:3006) coordinate reference system. All vector data was rasterized with resolution 10 m to be processed and used in the analyses. Environmental variables, data sources, and data processing are described in Table A2.

Layer	Description	Image type	Year	Original resolution	Institution	Source
Land use and land cover	National Land Cover Database (NMD). Land cover map over Sweden with 25 thematic classes. Classified using Snetinel-2 satellite images and LIDAR data as well as existing national maps like soil maps, hydography, roads and buildings. The original classes are: 111 – Pine forest not on wetlands, 112 – Spruce forest not on wetlands, 113 – Mixed coniferous forest not on wetlands, 114 – Broadleaved forest not on wetlands, 115 – Mixed forest not on wetlands, 116 – Temporarily non-forest (clear-cuts etc.) not on wetlands, 121 – Pine forest on wetlands, 122 – Spruce forest on wetlands, 123 – Mixed coniferous forest on wetlands, 124 – Broadleaved forest on wetlands, 125 – Mixed forest on wetlands, 126 – Temporarily non-forest (clear-cuts etc.) on wetlands, 211 – Arable land, 212 – Grassland, 221 – Artificial surfaces, 231 – Bare rock and other areas with sparse vegetation, 241 – Open wetlands, 242 – Forested wetlands, 251 – Natural grasslands, 252 – Alpine heaths, 253 – Alpine meadows, 261 – Glaciers and permanent snow, 271 – Inland water, 272 – Marine water. The classes were re-classified as: deciduous forest (114, 124), coniferous forest (111, 112, 113, 115, 121, 122, 123), open areas (116, 211, 212, 231, 251, 252, 253), mires (125, 241, 242), and anthropogenic areas (221). Classes 261 and 272 were not present in the study area, and class 271 was removed because few GPS position were found within water. Class 126 was updated for each year based on clear cut data.	Raster	2018	10 m	Swedish Environmental Protection Agency (Naturvårdsverket)	[1]
Clear cuts	Final fellings data from the Forest Attribute Map (Digital Open Forest Data), consisting of polygons and corresponding year of cut. Clear cuts year was compared with the acquisition time of GPS positions and land use classes were updated for those positions when the acquisition time occurred after the clear cut year.	Vector	2020	NA	The Swedish Forest Agency	[2]
Start of season (SOS), end of season (EOS), and instantaneous rate of green-up (IRG)	Derived from Normalized Difference Vegetation Index (NDVI) downloaded from Google Earth Engine (MOD13Q1.061 Terra Vegetation Indices 16-Day Global 250 m) using the R package phenofit (https://github.com/eco-hydro/phenofit).	Raster	2022	250 m	Google Earth Engine	[3]
Elevation	Terrain model of Sweden	Raster	NA	1 m, resampled to 10 m	Lantmäteriet (Swedish mapping, cadastral and land registration authority)	[4]

(continued)

Layer	Description	Image type	Year	Original resolution	Institution	Source
Slope, aspect	Derived from elevation using the module r.slope.aspect in GRASS GIS 7.8.	Raster	NA	10 m	Computed for this study	Derived data
Terrain position index (TPI)	Computed based on the elevation, with neighborhood sizes 150 m, 250 m, and 510 m. TPI was reclassified following the approach from Weiss (2001) into six categories: valley, lower slope, flat, middle slope, upper slope, ridge. Reference: Weiss, A.D., 2001. Topographic Position and Landforms Analysis. ESRI User Conference.	NA	NA	10 m	Computed for this study	Derived data
Wind turbines	Position for each wind turbine, not used directly in the analysis	Vector	2019	NA	County administrative boards and Swedish Energy Agency	[5]
Zone of influence (ZOI) of wind turbines	Either the density of wind turbines of and the decay-distance to the nearest wind turbine, computed at scales 100 m, 250 m, 500 m, 1 km, 2.5 km, 5 km, and 10 km. ZOIs were computed using an exponential distance decay shape using the oneimpact package in R (https://github.com/NINAnor/oneimpact/).	NA	2019	10 m	Computed for this study	Derived data
Mines	Active mining areas	Vector	NA	NA	The Geological Survey of Sweden (SGU)	[6]
Zone of influence (ZOI) of mine	Computed considering an exponential decay distance to the mine, with ZOI radius 100 m, 250 m, 500 m, 1 km, 2.5 km, 5 km, and 10 km. ZOIs were computed using the oneimpact package in R (https://github.com/NINAnor/oneimpact/).	NA	NA	10 m	Computed for this study	Derived data
Public roads	Public (main) roads. Used to compute the (log) distance to the nearest feature, used in the statistical models.	Vector	2019	NA	Lantmäteriet (Swedish mapping, cadastral and land registration authority)	[7]
Private roads	Private roads (mostly private and forestry roads). Used to compute the (log) distance to the nearest feature as input for the statistical models.	Vector	2019	NA	Lantmäteriet (Swedish mapping, cadastral and land registration authority)	[7]
Trails	Tourist trails. Used to compute the (log) distance to the nearest feature as input for the statistical models.	Vector	2019	NA	Lantmäteriet (Swedish mapping, cadastral and land registration authority)	[7]

(continued)

Layer	Description	Image type	Year	Original resolution	Institution	Source
Power lines	Power lines. Used to compute the (log) distance to the nearest feature as input for the statistical models.	Vector	2019	NA	Lantmäteriet (Swedish mapping, cadastral and land registration authority)	[7]
Houses	Houses and associated building. Used to compute the (log) distance to the nearest feature as input for the statistical models.	Vector	2019	NA	Lantmäteriet (Swedish mapping, cadastral and land registration authority)	[7]

Table A2. Description of the environmental variables used for estimating reindeer habitat selection and movement patterns.

3 Summary statistics of environmental variables

Variable	Average (range) or proportion
Elevation (m)	370 (231 - 648)
Slope (degrees)	3 (0 - 63)
Start of season (doy)	136 (103 - 247)
End of season (doy)	302 (259 - 331)
Distance to public roads (m)	2107 (0 - 10890)
Distance to private roads (m)	453 (0 - 4924)
Distance to wind turbines (m)	21928 (0 - 72341)
Distance to power lines (m)	6893 (0 - 22152)
Deciduous forests	14.46%
Open lands	20.01%
Anthropogenic areas	1.49%
Coniferous forests	43.12%
Mires	20.92%

Table A3. Summary statistics (mean and range, or proportion of land use classes) of the environmental variables used for estimating reindeer habitat selection and movement patterns.

4 Data source references

[1] https://www.naturvardsverket.se/en/services-and-permits/maps-and-map-services/national-land-cover-database/

[2] https://www.skogsstyrelsen.se/en/digital-open-forest-data/

[3] https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MOD13Q1

[4] https://www.lantmateriet.se/en/geodata/our-products/product-list/elevation-model-download/

[5] https://ext-webbgis.lansstyrelsen.se/vbk/

[6] https://apps.sgu.se/kartvisare/kartvisare-mineralrattigheter.html

[7] https://www.lantmateriet.se/en/geodata/our-products/product-list/topography-10-download-vector/

5 References

GRASS Development Team. (2017). Geographic Resources Analysis Support System (GRASS GIS) Software, Version 7.8. Open Source Geospatial Foundation. http://grass.osgeo.org